The endpoint case for two open intervals

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Theorem 1. *Let $A \subset \mathbb{T}$ be a union of two non-empty disjoint open intervals with*

$$\mu(A) = \frac{1}{5}.$$

If

$$A + A - 3 \cdot A \neq \mathbb{T},$$

then, up to translating A and replacing A by $-A$, we have

$$A = \left(0, \frac{1}{10}\right) \cup \left(\frac{1}{2}, \frac{3}{5}\right).$$

In particular, $2 \cdot A$ is contained in an interval of length $\frac{1}{5}$.

Proof. The property of $A+A-3\cdot A$ being equal to $\mathbb{T}$ is unchanged by translating A or by replacing A by $-A$. As in the closed-interval proof, we may therefore write

$$A = I \cup J, \qquad I = (0, \alpha), \qquad J = (x, x+\beta),$$

where

$$0 < \alpha < x < x + \beta < 1, \tag{1}$$

$$\alpha \geq \beta, \tag{2}$$

$$x - \alpha \leq 1 - (x + \beta). \tag{3}$$

Since $\mu(A) = \frac{1}{5}$, we have

$$\alpha + \beta = \frac{1}{5}.$$

Condition (3) is equivalent to

$$x \leq \frac{1 + \alpha - \beta}{2} = 3\alpha + 2\beta. \tag{4}$$

We will show that, unless

$$\alpha = \beta = \frac{1}{10}, \qquad x = \frac{1}{2}, \tag{5}$$

we have $A + A - 3 \cdot A = \mathbb{T}$.

Let

$$\begin{aligned}
R_1 &= I + I - 3 \cdot I = (-3\alpha, 2\alpha), \\
R_2 &= I + J - 3 \cdot I = (x - 3\alpha, x + \alpha + \beta), \\
R_3 &= J + J - 3 \cdot I = (2x - 3\alpha, 2x + 2\beta), \\
R_4 &= I + I - 3 \cdot J = (-3x - 3\beta, -3x + 2\alpha), \\
R_5 &= I + J - 3 \cdot J = (-2x - 3\beta, \alpha + \beta - 2x), \\
R_6 &= J + J - 3 \cdot J = (-x - 3\beta, -x + 2\beta).
\end{aligned}$$

All of these intervals have length strictly smaller than 1, so we use them in proper form: a point $y \in \mathbb{T}$ belongs to one of these intervals if and only if, for some $k \in \mathbb{Z}$, the real number $y + k$ belongs to the displayed interval. Clearly each R_i is contained in $A + A - 3 \cdot A$.

We first record the strict inequalities that replace the non-strict endpoint inequalities in the closed-interval proof.

Claim 1. *If* (5) *does not hold, then*

$$x < 5\alpha \qquad \text{and} \qquad x < 4\alpha + \beta.$$

Proof. By (4) and $\alpha \geq \beta$,

$$x \leq 3\alpha + 2\beta \leq 5\alpha.$$

If equality held in $x \leq 5\alpha$, then we would need $\alpha = \beta$ and $x = 3\alpha + 2\beta$. Since $\alpha + \beta = \frac{1}{5}$, this gives $\alpha = \beta = \frac{1}{10}$ and $x = \frac{1}{2}$, which is exactly (5). Hence, outside the exceptional case, $x < 5\alpha$.

Similarly,

$$x \leq 3\alpha + 2\beta \leq 4\alpha + \beta,$$

and equality again forces $\alpha = \beta = \frac{1}{10}$ and $x = \frac{1}{2}$. Thus $x < 4\alpha + \beta$ outside the exceptional case. $\square$

Claim 2. *Assume* (5) *does not hold. Then*

$$R_1 \cup R_2 \cup R_3$$

covers the interval

$$[0, 2x + 2\beta)$$

and also covers the interval $(1 - 3\alpha, 1)$.

Proof. Modulo 1, the interval $R_1 = (-3\alpha, 2\alpha)$ covers

$$[0, 2\alpha) \cup (1 - 3\alpha, 1).$$

By Claim 1, $x < 5\alpha$, so

$$x - 3\alpha < 2\alpha.$$

Also $2\alpha < x + \alpha + \beta$, since $\alpha < x + \beta$. Hence R_2 contains 2α, and therefore $R_1 \cup R_2$ covers $[0, x + \alpha + \beta)$.

Again by Claim 1, $x < 4\alpha + \beta$, so

$$2x - 3\alpha < x + \alpha + \beta.$$

Also $x + \alpha + \beta < 2x + 2\beta$, since $\alpha < x + \beta$. Hence R_3 contains $x + \alpha + \beta$, and $R_1 \cup R_2 \cup R_3$ covers $[0, 2x + 2\beta)$. □

It remains to cover the possible leftover interval

$$[2x + 2\beta, 1 - 3\alpha]. \tag{6}$$

If this interval is empty, then we are done. Thus assume from now on that it is non-empty. Then

$$2x + 2\beta \leq 1 - 3\alpha, \tag{7}$$

or equivalently, using $\alpha + \beta = \frac{1}{5}$,

$$2x \leq 2\alpha + 3\beta. \tag{8}$$

We now follow the original proof and try to cover the leftover using $R_4 \cup R_5$. The only difference is that the intervals are open, so one endpoint case survives.

Claim 3. *The intervals R_4 and R_5 cover the leftover interval* (6)*, except possibly in the case*

$$\alpha = \beta = \frac{1}{10}, \qquad x = \frac{1}{4}. \tag{9}$$

In that case, the only point of (6) *not covered by $R_4 \cup R_5$ is $\frac{7}{10}$.*

Proof. We check the leftover interval after shifting by -1. Thus we want to cover

$$[2x + 2\beta - 1, -3\alpha].$$

The intervals R_4 and R_5 overlap, because

$$-2x - 3\beta < -3x + 2\alpha$$

is equivalent to $x < 2\alpha + 3\beta$, which follows from (8). Therefore

$$R_4 \cup R_5 = (-3x - 3\beta, \alpha + \beta - 2x).$$

The left endpoint is strictly covered, since

$$-3x - 3\beta < 2x + 2\beta - 1$$

is equivalent to $x + \beta > \frac{1}{5}$, which follows from $x > \alpha$ and $\alpha + \beta = \frac{1}{5}$.

For the right endpoint, we need

$$-3\alpha < \alpha + \beta - 2x,$$

which is equivalent to

$$2x < 4\alpha + \beta.$$

By (8) and $\alpha \geq \beta$,

$$2x \leq 2\alpha + 3\beta \leq 4\alpha + \beta.$$

Thus the desired strict inequality can fail only if both inequalities are equalities. This forces $\alpha = \beta = \frac{1}{10}$ and $2x = 2\alpha + 3\beta = \frac{1}{2}$, i.e. $x = \frac{1}{4}$.

In that case, the leftover interval (6) is the singleton

$$\left\{ \frac{7}{10} \right\},$$

and $\frac{7}{10} - 1 = -\frac{3}{10}$ is exactly the right endpoint of R_5, so it is not covered by $R_4 \cup R_5$. $\square$

This is the only place where the patch by R_6 is needed. In the fake case (9), we have

$$R_6 = \left(-\frac{11}{20}, -\frac{1}{20} \right),$$

and

$$\frac{7}{10} - 1 = -\frac{3}{10} \in R_6.$$

Therefore the apparent missing endpoint is covered by $R_6 = J + J - 3 \cdot J$.

We have shown that, outside the exceptional case (5), the union of the intervals $R_1, \ldots, R_6$ covers all of $\mathbb{T}$. Hence $A + A - 3 \cdot A = \mathbb{T}$ outside (5). Therefore, if $A + A - 3 \cdot A \neq \mathbb{T}$, the exceptional case (5) must hold.

Finally, in the exceptional case,

$$A = \left(0, \frac{1}{10} \right) \cup \left(\frac{1}{2}, \frac{3}{5} \right),$$

and hence

$$2 \cdot A = \left(0, \frac{1}{5} \right),$$

which is contained in an interval of length $\frac{1}{5}$. A direct calculation also shows that this exceptional set really does not cover the circle: the six intervals above have union

$$\mathbb{T} \setminus \left\{ \frac{1}{5}, \frac{7}{10} \right\}.$$

This completes the proof. $\square$